

LECTURE TEN (193-248)

High-Dimensional Linear Regression

Topic 1: Linear Regression Models

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t$$

$$y_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 \cos \frac{2\pi t}{12} + \beta_4 \sin \frac{2\pi t}{12} + \varepsilon_t$$

(sm.OLS) $\hat{\beta}$ by least squares.

$$\left(\frac{1}{S(\beta)}\right)^{n/2}$$

Topic 2: Non linear Regression Models

$$\textcircled{f} \rightarrow y_t = \beta_0 + \beta_1 \cos 2\pi f t + \beta_2 \sin 2\pi f t + \varepsilon_t$$

$$y_t = \beta_0 + \beta_1 t + \beta_2 (t-c)_+ + \varepsilon_t$$

c unknown

Again inference by least squares.

$$\min_{\beta_0, \beta_1, \beta_2} \sum_{t=1}^n [y_t - \beta_0 - \beta_1 t - \beta_2 (t-c)_+]^2$$

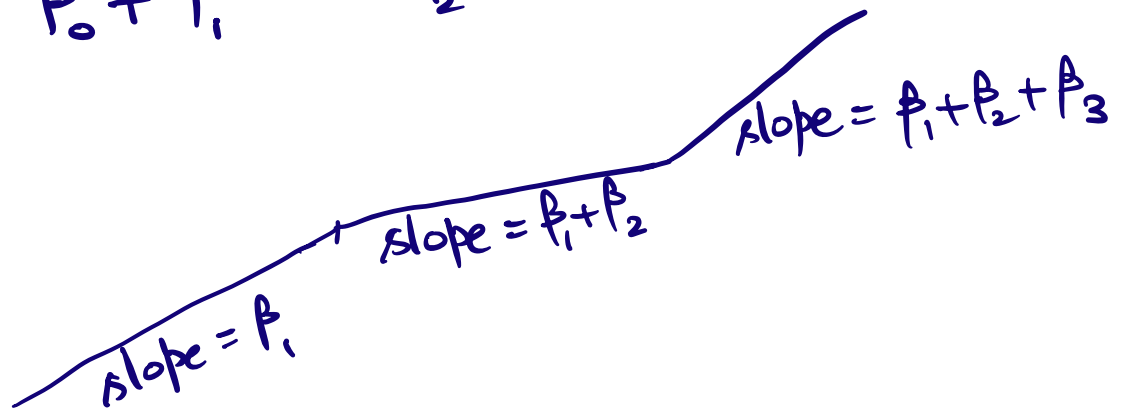
$\hat{c}$ minimizer of $RSS(c)$

$$\boxed{RSS(c)} = \min_{\beta_0, \beta_1, \beta_2} \sum_{t=1}^n [y_t - \beta_0 - \beta_1 t - \beta_2 (t-c)_+]^2$$

$$\left(\frac{1}{RSS(c)} \right)^{\frac{n-p}{2}} |X_c^T X_c|^{-1/2} I(c)$$

$$y_t = \beta_0 + \beta_1 t + \underbrace{\beta_2 (t - c)_+}_{\text{slope} = \beta_1 + \beta_2} + \varepsilon_t$$

$$y_t = \beta_0 + \beta_1 t + \beta_2 (t - c_1)_+ + \beta_3 (t - c_2)_+ + \varepsilon_t$$



c_1, c_2
 $\beta_0, \beta_1, \beta_2, \beta_3$

$RSS(c_1, c_2)$

 $\rightarrow 100^2 \times n$

$$\beta_0 + \beta_1 t + \beta_2 (t - c_1)_+ + \beta_3 (t - c_2)_+ + \beta_4 (t - c_3)_+ + \varepsilon_t$$

$RSS(c_1, c_2, c_3)$

Estimation Complexity will increase with k .

$$RSS(c_1, \dots, c_k)$$

Alternative Approach

$$\sum_{t=1}^n \left[y_t - \beta_0 - \beta_1 t - \beta_2 (t - c_1)_+ - \dots - \beta_{k+1} (t - c_k)_+ \right]^2$$

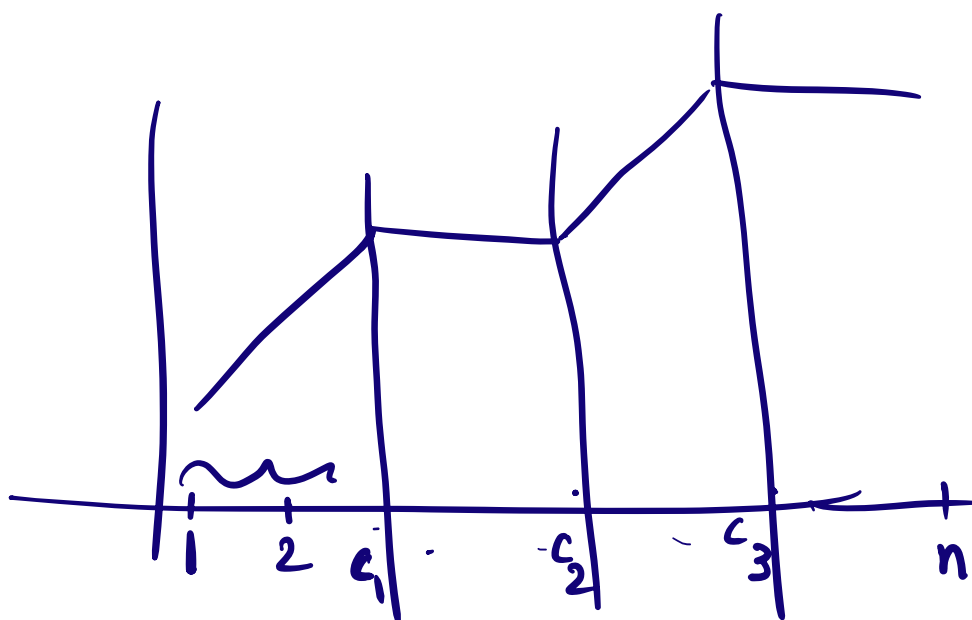
$$F(\beta_0 \dots \beta_{k+1}, c_1 \dots c_k) \quad \text{optimize over } \beta_0, \dots, \beta_{k+1}, c_1, \dots, c_k$$

PyTorch ✓

Initialize $\beta_0 \dots \beta_{k+1}, c_1 \dots c_k$

If $(t - c_j)_+$ is replaced by $I(t > c_j)$

Gradient Descent.



Fix $c_1 \dots c_k$ as quantiles of $x_{(1), \dots, (n)}$

Then estimate $\beta_0 \dots \beta_{k+1}$

Initialization.

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c_1) + \dots + \beta_{k+1} \text{ReLU}(t - c_k) + \varepsilon_t$$

$k = 5086$

- ① Choice of k is tricky.
- ② Parameter Estimation is non-trivial.
- ③ If k is not very small, there is risk of overfitting.

→ Fix either by Regularization or by using informative Bayesian Priors.

NOT LEAST SQUARES.

Model (with large k)

We will consider the extreme version of

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - c_1) + \dots + \beta_{k+1} \text{ReLU}(t - c_k) + \varepsilon_t$$



New Model:

$$y_t = \beta_0 + \beta_1 t + \beta_2 \text{ReLU}(t - 2) + \beta_3 \text{ReLU}(t - 3) + \beta_4 \text{ReLU}(t - 4) + \dots + \beta_{n-1} \text{ReLU}(t - (n-1)) + \varepsilon_t$$

$$\text{ReLU}(t - 5.7) = 0.3 \text{ReLU}(t - 5) + 0.7 \text{ReLU}(t - 6)$$

→ for all $t \leq 5$

Δ all $t \geq 6$

$t \leq 5$ both sides = 0

$t \geq 6$ lhs : $t - 5.7$

rhs : $(0.3)(t-5) + 0.7(t-6)$
 $= t - 5.7$

$$y_t = \beta_0 + \beta_1 t + \beta_2 (t-2)_+ + \dots + \beta_{n-1} (t-(n-1))_+ + \varepsilon_t$$

Comments

① Linear Model but high-dimensional with n regression coefficients

② Least Squares will NOT work.

(Perfect Overfitting
Δ so we will reproduce the data).

Reason : We can always choose $\beta_0, \beta_1, \dots, \beta_{n-1}$ so that

$$y_t = \beta_0 + \beta_1 t + \beta_2 (t-2)_+ + \dots + \beta_{n-1} (t-(n-1))_+ + \varepsilon_t$$

for all $t = 1, \dots, n$

Here are the explicit values of β 's.

Slight change in model notation

$$y_t = \beta_0 + \beta_1 (t-1) + \beta_2 (t-2)_+ + \dots + \beta_{n-1} (t-(n-1))_+ + \varepsilon_t$$

$$\sum_{t=1}^n \left[y_t - \beta_0 - \beta_1(t-1) - \dots - \beta_{n-1}(t-(n-1))_+ \right]^2$$

$$y_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \dots + \beta_{n-1}(t-(n-1))_+$$

choose β_j 's so that this is true for all $t=1, \dots, n$

$$(t=1) \quad y_1 = \beta_0$$

$$(t=2) \quad y_2 = \beta_0 + \beta_1 \Rightarrow \beta_1 = y_2 - \beta_0 = y_2 - y_1$$

$$(t=3) \quad y_3 = \beta_0 + 2\beta_1 + \beta_2 \Rightarrow \beta_2 = y_3 - 2\beta_1 + \beta_0$$

$$\Rightarrow \beta_2 = \underbrace{(y_3 - y_2)} - \underbrace{(y_2 - y_1)}$$

More generally,

$$\beta_j = (y_{j+1} - y_j) - (y_j - y_{j-1})$$

Next lecture: Different Estimation Strategy

$$y_t = \beta_0 + \beta_1(t-1) + \sum_{j=2}^{n-1} \beta_j \text{ReLU}(t-j) + \epsilon_t$$

① Frequentist $\rightarrow$ Regularization

② Bayesian $\left(\sum \left(y_t - \beta_0 - \beta_1(t-1) + \sum_{j=2}^{n-1} \beta_j \text{ReLU}(t-j) \right)^2 + \lambda \text{Penalty} \right)$

$\beta_j \text{ iid } \text{Unif}(-C, C)$

↓
change to a more informative prior.